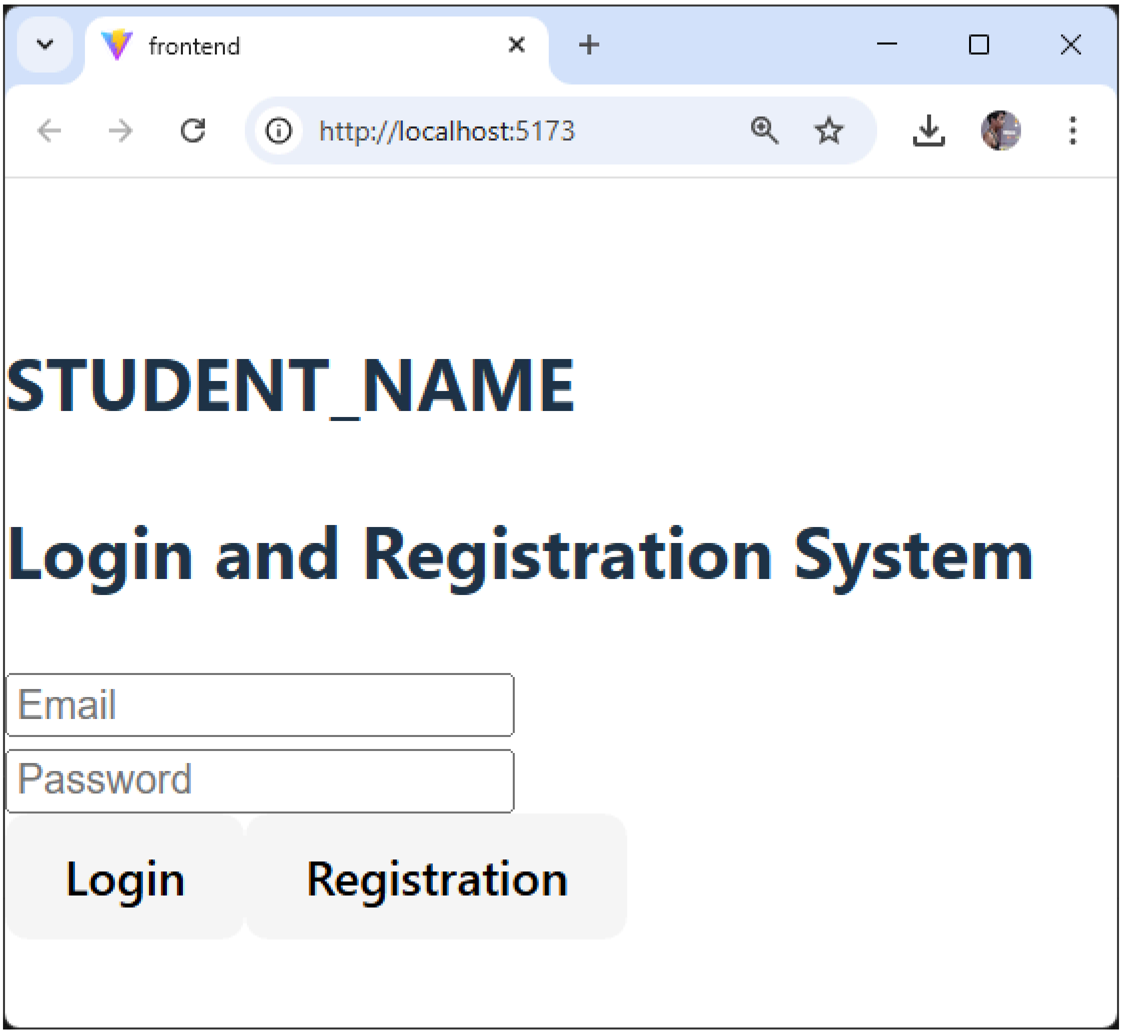
**IERG3840 Web Application Development Project (2025-26 Term 1)**
**Lab 3 Specification**

Lab 3 is to design a login and registration system with a MongoDB Atlas using React JS and Node.JS in your laptop. After reproducing the login system in the lecture note, you are going to implement your registration system. It will take 15% of the course total.

## Part 1: Demonstration (7% + Bonus 1%)

Date: 16th Oct, 2025 (Thur)
Time and Venue: In Class
You need to bring your own laptop for demonstration



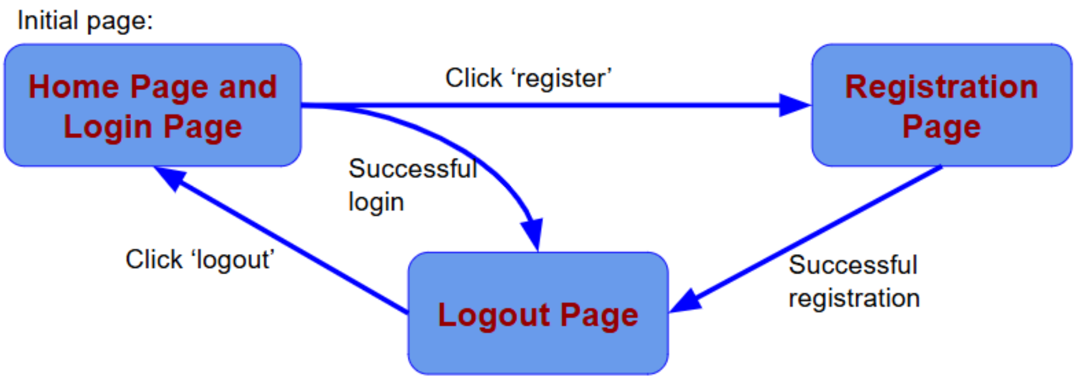
In your demonstration, you need to demonstrate the following features:

- Show your full name (0.5%)

- Reproduce the login system such that the login information is checked against with the data in your MongoDB. The two error messages, “User does not exist” and “Incorrect password”, should be shown properly. (1%)

- After successful login, you should redirect to the logout page, showing a logout button. If you click the logout button, you will redirect back to the login page. (1%)

- Create your login and registration system based on the following transition diagram. (1%)




- After clicking the register button in the home / login page, the web service redirects the user to the registration system. In the registration system, user login email, password and confirmed password are compulsory features. You need to check: (1.5%)
    - The user login email is in email format.
    - The password is at least 8 characters and at most 20 characters.
    - The password and the confirmed password are identical.

  You also need to display the error message properly, before the user submits the registration information.

- Implement one more feature in the registration page based on the last digit of your student ID. (1%)

| Last digit of your student ID | Student ID example | Feature | Error checking and/or implementation details |
|---|---|---|---|
| 0 | 1155000000 | Personal description | Non-empty with at most 100 characters |
| 1 | 1155000001 | Telephone number | Valid Hong Kong telephone format, and digits only |
| 2 | 1155000002 | Occupation | A dropdown menu with at least 8 choices |
| 3 | 1155000003 | Gender | Radio buttons with 3 exclusive choices, male, female and not specified |
| 4 | 1155000004 | Date of birth | Selection of date from calendar |
| 5 | 1155000005 | Favourite color | Selection of color from color picker |
| 6 | 1155000006 | Height and weight | Valid range with height in cm, weight in kg |
| 7 | 1155000007 | University | A dropdown menu with at least 8 choices |
| 8 | 1155000008 | Nickname | Non-empty with at most 20 characters |
| 9 | 1155000009 | District | Radio buttons with 3 exclusive choices, New Territories, Kowloon and Hong Kong |

- After submitting the information from the registration page, the server needs to check whether the account exists in your user collection in MongoDB or not. If yes, the error message, “account is already created”, should be displayed. If not, a new user document is inserted to your user collection in MongoDB. The document includes user email, password and your assigned feature. Password encryption is not required. (1%)

- [This task is challenging] Verify whether the user email is valid or not. (Bonus 1%):
  - Add a “verify” button in the registration page
  - After clicking the “verify” button, send a verification email from the server to the user email
  - Before completing the verification, an error message, “email is not verified”, is displayed

## Part 2: Question and Answer (5%)

You will be asked one set of problems out of the five sets below randomly after completing the demonstration. You need to answer the problem set verbally.
You can prepare a cheat sheet for the Q&A session, however, you cannot search online or search from the lecture notes during the Q&A session.
There are five points in each set.

Set 1:

(a) What is SQL? (1%)
(b) What is RDBMS? (1%)
(c) What is an ER diagram? (1%)
(d) What is an entity in SQL? (1%)
(e) What is an attribute in SQL? (1%)

Set 2:

(a) What is noSQL? (1%)
(b) What is a document in noSQL? (1%)
(c) What is a collection in noSQL? (1%)
(d) What is a cluster in MongoDB? (1%)
(e) What is the actual data format stored inside MongoDB? (1%)

Set 3:

(a) Suggest two major benefits of SQL over noSQL? (2%)
(b) Suggest two major benefits of noSQL over SQL? (2%)
(c) Suggest and explain one major reason why noSQL is getting more and more popular nowadays? (1%)

Set 4:

(a) What are the two major ways or directions of using MongoDB? (2%)
(b) What are CRUD operations? (2%)
(c) Do we need to create the collection in MongoDB? Why? (1%)

Set 5:

(a) In addition to String and Number, suggest four data types supported by MongoDB documents. (2%)
(b) What is the role of _id in MongoDB documents? (1%)
(c) Explain the following two queries. (2%)

```
// Query 1
db.students.find({ department: "IE" })
// Query 2
db.students.find().sort({ exam_score: -1 }).limit(1)
```

## Part 3: Short Report (3%)

Use the template, fill in all the necessary information and submit the **PDF** format on or before 17th Oct, 2025 (Fri). Without demonstration, no scores will be given for the submitted report.

## Late Policy

The following behaviours are regarded as late:

- Demonstration and Q&A are not performed on the specified date without justification
- Short report is submitted after the deadline

Late penalties will be counted on the late parts only.

- Demonstration and Q&A: 30% off for late within one week
- Report: 30% off for late within one week
- No scores after one week

**< End of Specification >**